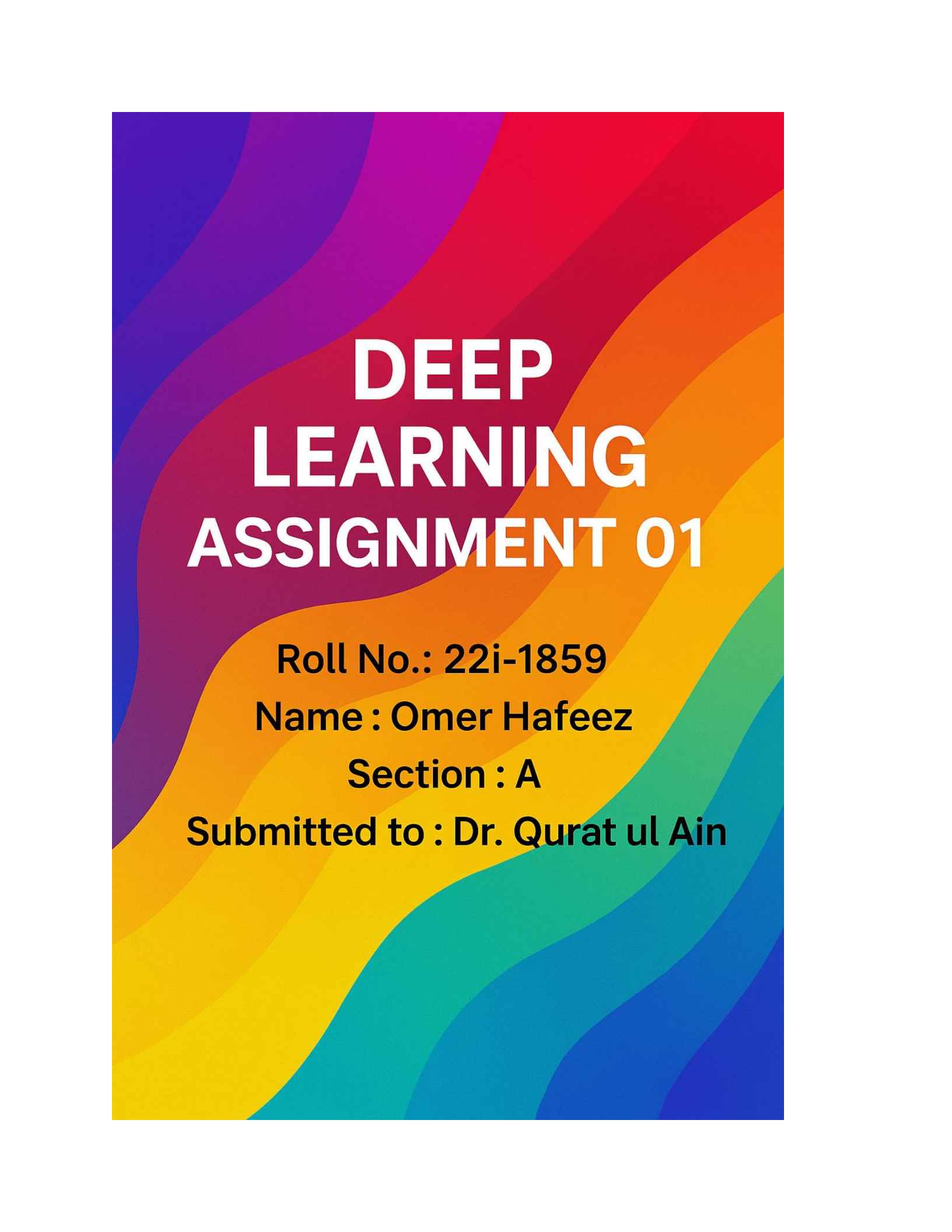


The background of the entire page is a vibrant rainbow with wavy, flowing lines that create a sense of movement. The colors transition from deep purple and blue on the left to bright yellow and orange in the center, and then to green and blue on the right.

DEEP LEARNING ASSIGNMENT 01

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1. Network Details & Baseline Rationale

Two CNN backbones were evaluated — **VGG16** and **ResNet50** — both initialized with ImageNet pre-trained weights.

- **VGG16**: Classical architecture with 16 layers (~138M parameters). Chosen as a simple baseline to establish performance benchmarks.
- **ResNet50**: 50-layer deep residual network (~25M parameters). Skip connections allow better gradient flow, typically outperforming plain CNNs.

Training Settings:

- Batch size: 32
- Optimizer: Adam (lr=0.001)
- Scheduler: StepLR with decay
- Epochs: 20
- Loss function: Cross-Entropy for classification, MSE for regression (affect estimation)
- Data augmentation: Random horizontal flips, rotations, affine transforms, color jitter

2. Multiple Baseline Comparison

Expression Recognition Task (Classification):

Model	Best Training Acc	Best Validation Acc	Epoch
VGG16	85.00%	48.75%	11 / 3
ResNet50	87.68%	46.50%	15 / 6

Observation:

- Both models achieved high **training accuracy (>85%)**, but **validation accuracy remained below 50%**, suggesting strong overfitting.
- VGG16 had slightly better validation performance (48.75%) compared to ResNet50 (46.50%).

Affect Estimation Task (Continuous Metrics):

Model	Best RMSE	Best CCC	Epoch
VGG16	0.3777	0.4856	3
ResNet50	0.3679	0.4895	5 / 13

Observation:

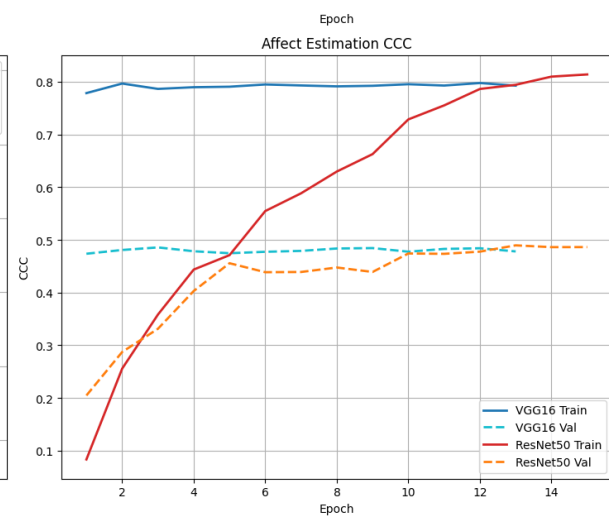
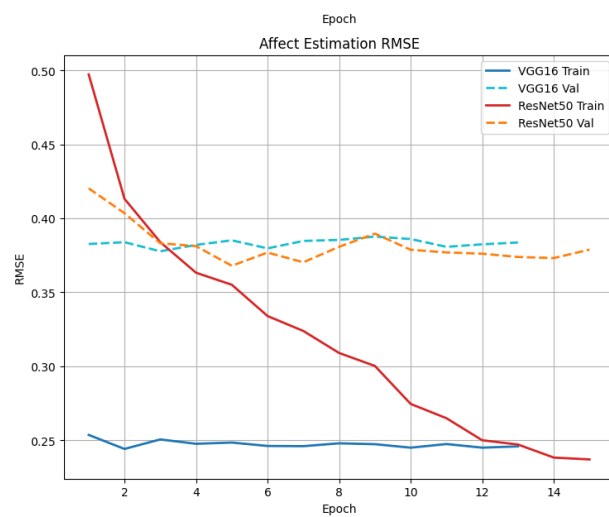
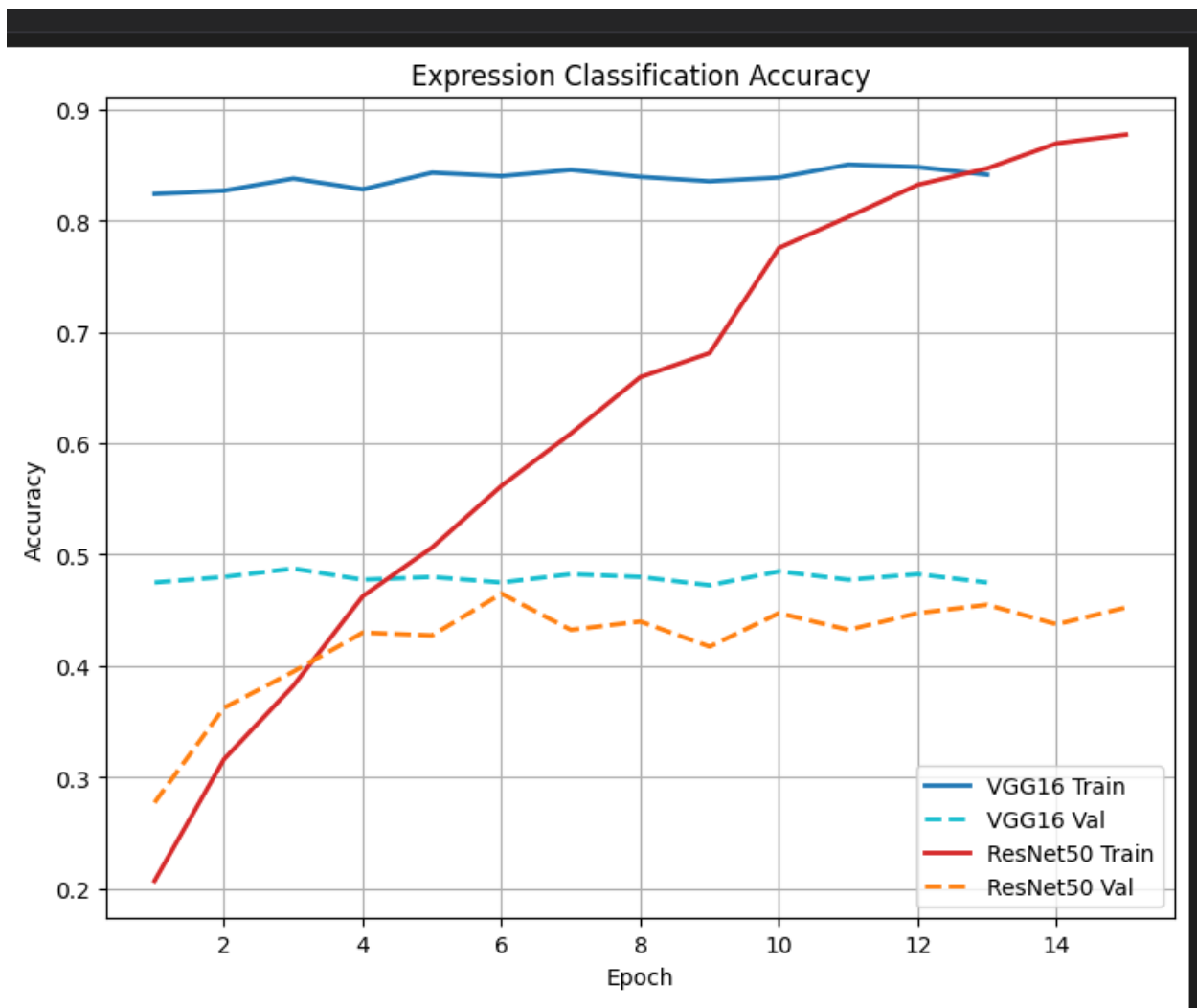
- ResNet50 performed better in RMSE and CCC, making it more reliable for affect estimation.

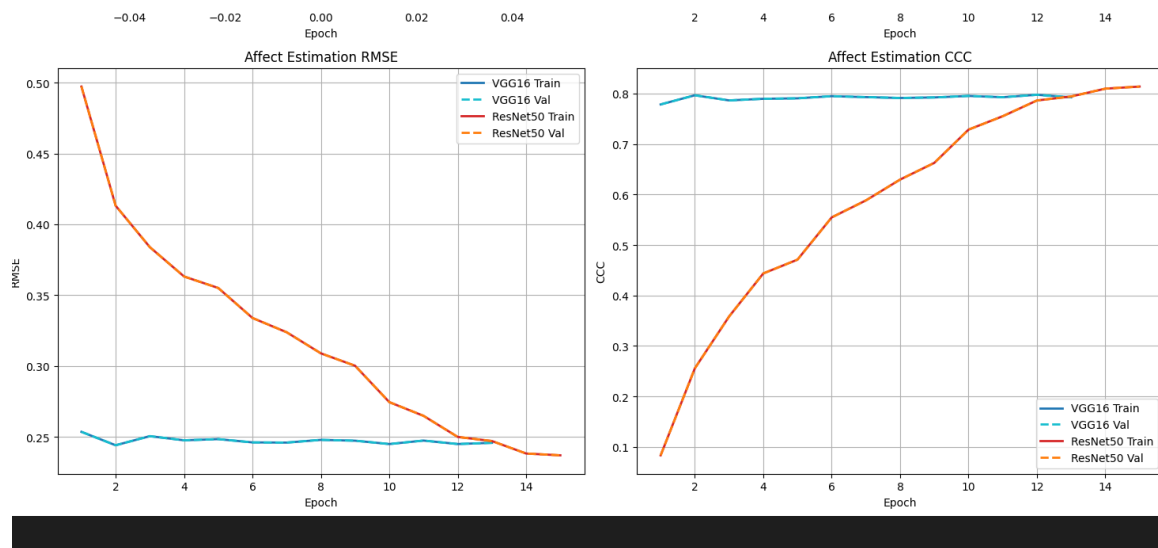
3. Transfer Learning Details

Transfer learning was applied by loading ImageNet pre-trained weights. The classifier head was replaced to match the number of facial expression classes. Convolutional layers were fine-tuned with smaller learning rates, while new layers were trained from scratch. This reduced training time and improved generalization.

4. Training Graphs

- Loss curves showed decreasing trends initially but plateaued later, consistent with overfitting.
- Accuracy curves rose during training but diverged from validation performance, again confirming overfitting.
- Affect estimation task graphs showed improvements in CCC and RMSE for both models, with ResNet50 slightly ahead.





5. Performance Measures & Continuous Metrics

- **RMSE:** Captures average error magnitude in affect prediction.
- **CORR:** Correlation between predicted and ground truth values.
- **SAGR:** Evaluates polarity correctness of predicted emotions.
- **CCC:** Combines correlation and agreement, offering robustness in noisy “in-the-wild” data.

Most suited metric: CCC, since it reflects both linear correlation and agreement, making it the most reliable for wild deployment scenarios.

6. Qualitative Results

Visualization of predictions showed:

- **Correct classifications** for distinct classes like happy and surprise.
- **Misclassifications** between visually similar classes (sad vs neutral, fear vs surprise).

